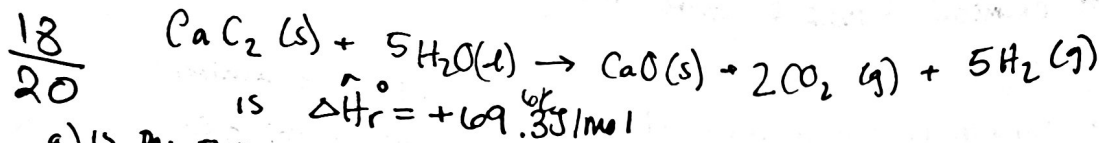


1) ~~The~~ Standard heat of rxn $\text{H}_2\text{O} \neq \text{H}_2\text{O}$ CITE 210



a) Is the rxn exothermic or endothermic @ 25°C. would you have to heat or cool the reactor to keep the temp. constant? what would the temp do if the reactor ran adiabatically? what can you infer about the E required to break the molec. bonds of the reactants & that released when the products bonds are formed?

Rxn is endothermic b/c ΔH_r° is positive.

You would have to heat the reactor to keep temp constant

Since $\Delta G = 0$, Temp will go down

You need a lot of energy (or more E) to break bonds of the reactants

However, you don't need as much E in products to form bonds, so you can have extra Energy

b.) Calculate ΔU_r for this rxn, explain briefly the physical significance of your calc. value.

$\Delta U_r \approx \Delta H_r - RT \Delta n_r$

(gas) $+69.35 \frac{\text{kJ}}{\text{mol}} - (8.314 \frac{\text{J}}{\text{mol K}}) (298.15 \text{ K}) (7)$
 $\frac{1}{1000} = 51.94 \frac{\text{kJ}}{\text{mol}}$

Stand. conditions

25°C & 1 atm

↓
298.15 K

$\Delta U_r = +51.94 \frac{\text{kJ}}{\text{mol}}$

✓ → Brief explanation? → Internal energy change WFO

(c.) Suppose you charge 150.0g of CaC_2 & liq. water into a rigid container @ 25°C

heat the container until the calcium carbide reacts completely & cool the products back down

to 25°C, condensing essentially all of the unconsumed H_2O . Write & simplify the E bal eq for closed constant-volume & use it to det. the net amount of heat (kJ) that must be transferred to (or from) the reactor. Indicate the direction of heat transfer.

$\Delta H + \Delta PE + \Delta KE = Q + W \quad Q = \Delta H = n \Delta H_r$

150.0g CaC_2
 2 water

$1 \text{ mol } \text{CaC}_2(\text{s}) \times \frac{64.1 \text{ g}}{1 \text{ mol } \text{CaC}_2}$

$\frac{\text{mass}}{\text{molar mass}} \frac{150 \text{ g } \text{CaC}_2}{64.1 \text{ g/mol}} = 2.34 \text{ moles} \times \text{amount of E absorbed} = 162.31 \text{ kJ}$
 $(69.35) \frac{\text{kJ}}{\text{mol}}$

↓
52.0 kJ/mol

$\text{MW} = 64.1 \text{ g}$
 CaC_2

1) 18

2) 20

3) 20

4) 18

5) 20

$\frac{a6}{100}$

2. Murphy 6.32 N.B Consider the change in E in the inner combustion chamber as being completely due to chemical rxn & ignore the sensible energy change.

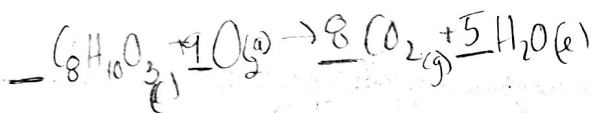
20
20 $C_8H_{10}O_3$ & its products.
In a bomb calorimeter a sample is placed into a thinner combustion chamber & combusted to CO_2 & H_2O . The combust. cham. sits inside another chamber which is filled w/ an exact amount of H_2O . Temp of the outer is carefully monitored.
In one experiment, 2.7 g of a proprietary organic chemical, $C_8H_{10}O_3$ is placed in a bomb calorimeter & combusted.

outer chamber = 650 g H_2O

$T_i = 23.6^\circ C$

$T_f = 36.2^\circ C$

calc. ΔH of formation (kJ/g)



$$\left(\frac{650g}{1}\right)(H_2O) \times \left(\frac{4.18 J}{1g^\circ C}\right) \times (36.2 - 23.6^\circ C) = -34,234 J = -34.234 kJ$$

$$Q_r = m \Delta U_r \rightarrow \Delta U_r = \frac{Q_r}{m} \rightarrow \frac{-34,234 J}{0.0175 mol} = -1,954,899.613 J/mol \rightarrow -1,954.9 kJ/mol$$

$$\left(\frac{2.7g}{1}\right) \left(\frac{C_8H_{10}O_3}{154.18g}\right) \times \left(\frac{1 mol}{154.18g}\right) \Delta H_r = \Delta U_r + RT \Delta n_r$$

$$-1,954.9 \frac{kJ}{mol} + \left(18.314 \times 10^{-3} \frac{kJ}{mol K}\right) (309.2 K) (1 mole)$$

$$\Delta H_r = -1,957.46 kJ/mol$$

$$\Delta H_r = \sum \nu_i H_{f,i} \quad \Delta H_{H_2O(l)} = -285.84 kJ/mol$$

$$\Delta H_{CO_2(g)} = -393.5 kJ/mol$$

$$-1,957.46 kJ = 8(-393.5 kJ/mol) + 5(-285.84) - \Delta H_{C_8H_{10}O_3} + \Delta H_{O_2(g)}$$

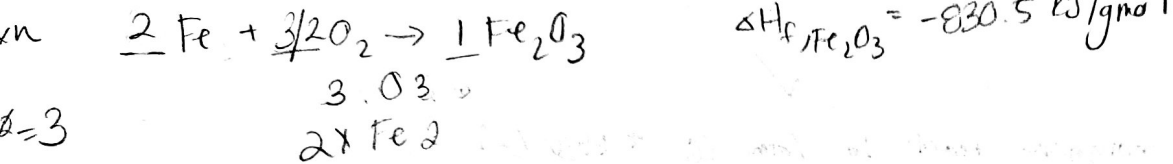
$$\Delta H_{C_8H_{10}O_3} = \left(\frac{2,619.74 kJ}{1 mol}\right) \left(\frac{1 mol}{154.18g}\right)$$

$$= -16.99 kJ/g$$

Q. 6.34

1 sized hot packs.... Fe fillings are placed in a sealed bag. When seal is broken
 air, air leaks in. O_2 reacts w/ iron producing iron oxides mainly Fe_2O_3
 res suggest that consumers want a slow heat release $\sim 100 \text{ kJ/h}$ & would like the
 packs to last 6 hrs. How much Fe would you put in the hot pack?
 how much mass of pack change over course of its use (total g & %).

$\frac{20}{20}$



re want $Q = 100 \text{ kJ/hr}$ for 6 hrs $Q = 600 \text{ kJ}$
 $\Delta \hat{H}_r^\circ = \Delta \hat{N}_r = -830.5 \text{ kJ/mol}$ $\frac{600 \text{ kJ}}{-830.5 \text{ kJ/mol}} = .722 \text{ moles Fe}_2\text{O}_3$

$\frac{.722 \text{ moles Fe}_2\text{O}_3}{1} \left(\frac{2 \text{ mol Fe}}{1 \text{ mol Fe}_2\text{O}_3} \right) \left(\frac{55.845 \text{ g Fe}}{1 \text{ mol Fe}} \right) = 80.6 \text{ g Fe}$

$\frac{.722 \text{ mol Fe}_2\text{O}_3}{1} \left(\frac{159.69 \text{ g}}{1 \text{ mol Fe}_2\text{O}_3} \right) = 115.3 \text{ g Fe}_2\text{O}_3$

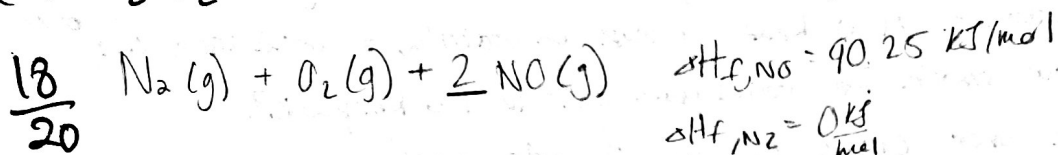
$115.3 \text{ g} - 80.6 = 34.70 \text{ g} \uparrow$

$\% \text{ change} = \frac{115.3 \text{ g} - 80.6 \text{ g}}{80.6} = 43.4\%$

1. Use tabulated heats of formation to work.

Following reactions in kJ/mol. Let stoich. coeff of the 1st reactant be one

(a) $N_2 + O_2$ react to form NO at in the gas state



$$\Delta H_f, N_2 = 0 \frac{\text{kJ}}{\text{mol}}$$

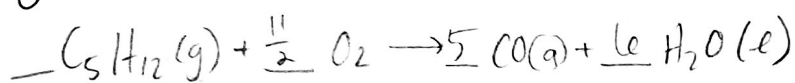
$$\Delta H_f, O_2 = 0 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta H_r = \sum \nu_i \Delta H_f$$

$$= 2(90.25 \frac{\text{kJ}}{\text{mol}}) - 0 \frac{\text{kJ}}{\text{mol}} - 0 \frac{\text{kJ}}{\text{mol}} = 180.5 \checkmark$$

(b.) gaseous n-pentane + oxygen react to form CO + H₂O (l)

$$\Delta H_f, C_5H_{12}(g) = -146.76 \frac{\text{kJ}}{\text{mol}}$$



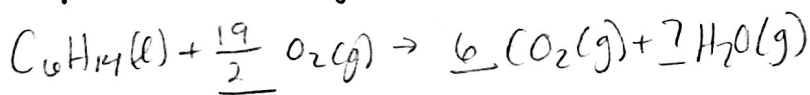
$$\Delta H_f, CO(g) = -110.53 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta H_f, H_2O(l) = -285.84 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta H_r = 5(-110.53) + 6(-285.84) - (-146.76 \text{ kJ/mol})$$

$$= -2120.93 \text{ kJ/mol} \checkmark$$

4c. liq n-hexane + oxy. $\rightarrow$ CO₂ + water vapor $\rightarrow$ use Hess's Law to derive formula



$$\Delta H_f, C_6H_{14}(l) = -198.9 \text{ kJ/mol}$$

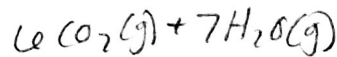
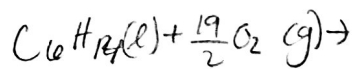
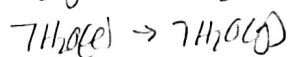
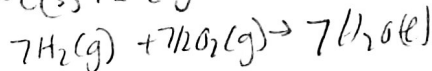
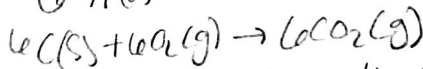
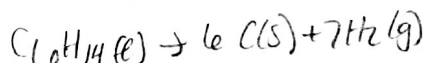
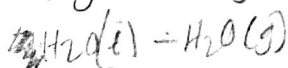
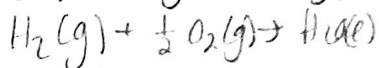
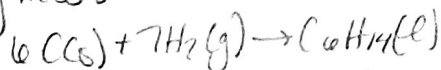
$$\Delta H_f, CO_2(g) = -393.5 \text{ kJ/mol}$$

$$\Delta H_f, H_2O(g) = -241.83 \text{ kJ/mol}$$

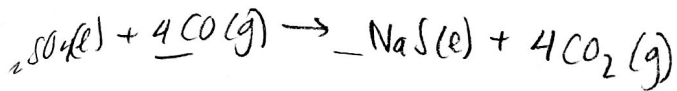
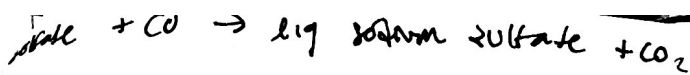
$$\Delta H_r = 6(-393.5) + 7(-241.83) - (-198.8 \text{ kJ/mol} - 0)$$

$$= -3855.01 \text{ kJ/mol} \checkmark$$

Using Hess's Law



$$6(-393.5) + 7(-241.83) - (-198.8) = -3885.01$$



$$\Delta H_r = [-323.94 + 4(-393.5) - (-1356.38) - 4(-110.53)] \text{ kJ/mol}$$

spiky 6.37

$$= -99.44 \text{ kJ/mol}$$

$$-138.2 \text{ kJ/mol}$$

pure sodium crystal (100g) sits in a steel cup

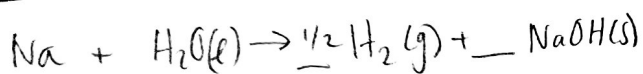
cup (1mL) H_2O is added $\text{H}_2(\text{g})$ produced.

What else happens? explain, as quantitative as possible

Why is it dangerous to add H_2O to a pre metal aka sodium.

20
20

Pure Na + 1mL H_2O



$$\left(\frac{100\text{g}}{22.98\text{g/mol}} \right) = .056 \text{ mol H}_2\text{O}$$

$$\left[4.35 - .056 \right]_{\text{Na}} \text{ mol} = 4.29 \text{ mol Na}$$

$$\left(\frac{100\text{g Na}}{22.98\text{g Na}} \right) = 4.35 \text{ mol Na}$$

$$\Delta H_r = [-425.9 - (-289.84) + 0 + 0] \frac{\text{kJ}}{\text{mol}} = -140 \frac{\text{kJ}}{\text{mol}}$$

$$\left(\frac{4.29 \text{ mol Na}}{1} \right) \left(\frac{22.98 \text{ g}}{1 \text{ mol Na}} \right) = 98.7 \text{ g Na}$$

$$c_p = 285 \text{ J/gmolC}$$

$$\frac{140 \text{ kJ}}{\text{mol}} (.056 \text{ mol}) + 4.29 \text{ mol} \left(\frac{285 \text{ J}}{\text{molC}} \right) \Delta T$$

$$\Delta T = 65^\circ \text{C}$$

The temp change is fairly large meaning even adding 1mL of H_2O - The Hydrogen gas being produced could possibly explode away because of the heated Na.

✓